

# EUGENE CHO

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## EDUCATION

University of California, Davis

Jun. 2026

B.S. in Computer Science

## EXPERIENCES

Sealing Technologies, a Parsons Company

May 2026 – Present

Software Engineer Intern

Columbia, MD

- Contributing to Zepharis AI, a fully air-gapped, full-stack local LLM platform that delivers context-aware reasoning, serving as a portable AI subject-matter expert for defensive cyber operators at the edge
- Building features across the stack that fuse LLM technology with operational tooling, enabling natural language querying, RAG over mission documentation, and unified command across security tools including Elastic, Vectra, and Security Onion

Kaiser Permanente

Jun. 2025 – Dec. 2025

Software Engineer Intern

Remote

- Built and optimized event-driven backend services in **Spring**, **Koa**, and **Kafka** to power an NLP-guided pre-appointment survey and smart triage system, streaming and processing over **500K** monthly unique sessions across patient intake workflows
- Led the end-to-end implementation of a School Note decisioning feature, introducing **SOAP(XML)**-based scaffold integration and employing a config-driven release agent pattern to improve scalability, observability, and long-term maintainability
- Resolved a **Redis** cache miss pattern in patient evisit lookup by introducing time-bucketed cache keys and negative caching with TTLs, reducing external API calls by over **99%** and cutting median latency by **95%**

Persist AI (YC W23)

Feb. 2026 – May 2026

Software Engineer Intern

Sacramento, CA

- Engineered a multi-agent web platform for pharmaceutical formulation workflows using **FastAPI**, **React**, and **Vite**, integrating **22** specialized backend agents into a unified system for project execution, collaboration, and research automation
- Built and maintained **37** backend API endpoints covering chat, session lifecycle, project CRUD, task management, job execution, snapshots, and visualization services, powering real-time agent-driven workflow orchestration and persistent project tracking
- Designed and supported real-time and asynchronous application patterns across **85+** Javascript and Python microservices, using WebSockets, SSE, background jobs, and multi-service orchestration to support long-running computations
- Added automated quality coverage across **190+** test files using **Playwright** and **pytest**, strengthening regression detection across UI, API, and end-to-end workflows

AggieWorks

Oct. 2024 – Jun. 2026

Software Engineer

Davis, CA

- Built [daviscattlelog.com](https://daviscattlelog.com), a full-stack course/professor insights platform with **React**, **FastAPI**, and **PostgreSQL** to help UC Davis students plan schedules and compare instructors, reaching over **40K** unique users and **6.3K+** monthly active users
- Refactored search API endpoints to return lightweight course IDs with vector embeddings and fuzzy matching, achieving **4×** faster retrieval by eliminating expensive data aggregation and reducing response payloads by **90%**
- Optimized frontend rendering with virtualization and backend performance with **redis caching** and **threadpool offloading**, cutting API latency by **90%** and redundant traffic by **99%**, improving site uptime from **88%** to **99.9%**

## PROJECTS

CollabRoomAI | SvelteKit, Typescript, TailwindCSS, Go (Fiber), WebSockets, OpenAI API

Nov. 2025

- Developed a real-time collaborative chat application with integrated AI agents using **Go/Fiber** backend and **SvelteKit** frontend, leveraging **WebSockets** for bidirectional messaging, supporting up to **500** concurrent users with **sub-50ms** latency
- Integrated **OpenAI API** to deploy multiple AI agents with distinct system prompts and configurable model selection, enabling automated real-time conversation analysis, summarization, and contextual responses within chat rooms
- Built a responsive chat UI with **Svelte 5**, **TypeScript**, and **Tailwind**, using custom reactive stores for room and user state and reusable components for chat and AI agent flows; delivered **40%** faster loads and **60%** component reuse across **5+** features

HangulStudy | React, TailwindCSS, Hangul, Sqlite3, Express.js

Sep. 2024

- Created an interactive Korean language learning platform with **React**, **Express.js**, and **SQLite**, featuring level-based practice sets (**100+** vocabulary words), a virtual Hangul keyboard, and a real-time scoring system to help users build Korean proficiency
- Wrote **RESTful API** endpoints to enable dynamic delivery across multiple difficulty tiers, serving **20+** students learning Korean

## TECHNICAL SKILLS

**Languages:** Python, Go, Java, C/C++, C#, Kotlin, SQL, JavaScript, TypeScript, PHP, HTML/CSS, R

**Frameworks & Libraries:** React.js, Next.js, Node.js, Express.js, Svelte, SvelteKit, FastAPI, Flask, Fiber, Koa, Django, Spring, Spring Boot

**Data, Messaging, & Cloud:** PostgreSQL, SQLite, MySQL, Oracle Database, Redis, Kafka, DigitalOcean, Oracle Cloud, Kubernetes

**Developer Tools:** Git, GitHub Actions, Docker, Linux/Unix, Bash, PostHog, Lens, Splunk, Dynatrace, Jenkins, Jest, Jtest, LaTeX

**Methodologies:** Scrum/Agile, Waterfall, Circular Development, DevOps, CI/CD, Object-Oriented Design, Unit Testing, Code Review

**Awards:** Eagle Scout Award, ACSL Intermediate Division Finalist, Ventura County Fire Department Citizen Award